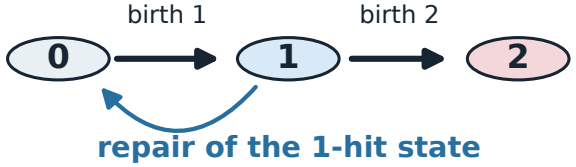
Repair filters mutational histories—not only mutational burden

A Same endpoint, two routes

Direct double hit · 1 event

Both alterations in one birth

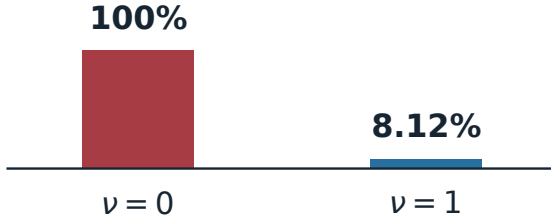
Sequential · 2 events



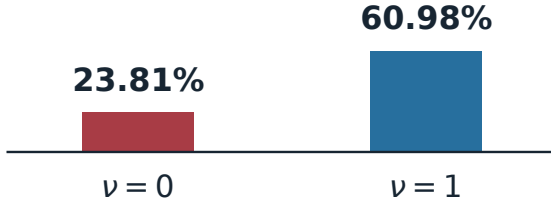
Sequential accumulation has an additional repairable intermediate state.

B Two-hit quantitative effect

Normalized two-hit abundance



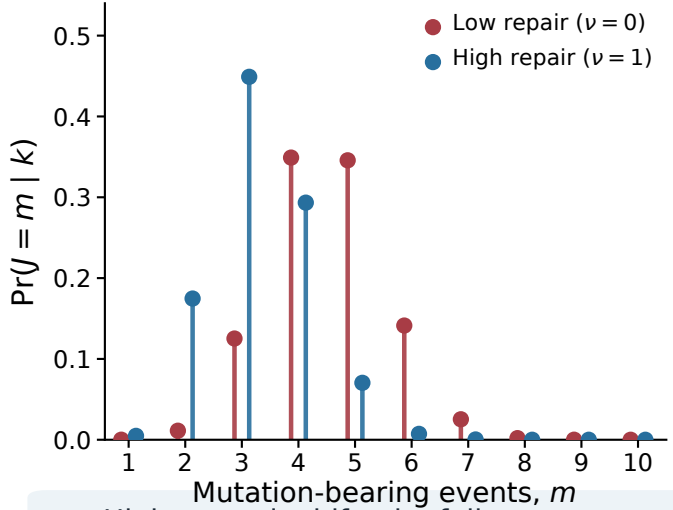
Direct-route share among two-hit cells



$$P_{\text{direct}}(\nu) = \frac{\lambda_1 + \nu}{\lambda_1 + \nu + 2\varepsilon r}$$

C General k-hit event ordering

Exact weighted routes: $k = 10$



Higher repair shifts the full event-count distribution toward fewer births.
MLR ordering when $g + r \geq 2$.

Parameters: $r = 0.8$, $g = 1.3$, $\varepsilon = 0.5$; $\nu \in \{0, 1\}$. All plotted values follow the rare-mutation route recursion.